Kept

Process Document

Tools, functional & non-functional decisions, and the visual evolution of the app

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Overview

Kept is a calm, privacy-first commitment app built on a single idea: hold one promise at a time, and treat a missed promise as a moment for reflection rather than punishment. This document records how it was built — the tools used across ideation, development, audio and testing; the functional decisions that define what the app does; the non-functional decisions that define how it behaves; and a visual record of how its central kept-promise visualization evolved.

Tools Used

Five tools spanned the work, from early ideation through cross-platform testing.

Tool	Stage	Role in the project
Claude (Anthropic)	Ideation & development	Design partner and pair-programmer: framing decisions against calm-technology and privacy principles, pressure-testing each feature for dark-pattern risk, and generating and iterating the React/Vite codebase — the logging, error-handling and canvas-acceleration infrastructure, the Living Artifact, and the successive iterations of the Signature visualization.
Google Antigravity	Ideation & development	Agent-first development platform (an IDE paired with an agent manager) used to plan, scaffold and iterate code across editor, terminal and browser — delegating multi-step implementation tasks and reviewing results through its artifacts (plans, diffs, screenshots).
ElevenLabs	Audio	Generated the ambient soundscape that the Living Artifact's Web Audio engine plays — the six mood regions of the audio bed, crossfaded with equal-power curves and tied to the app's state.
Microsoft Edge	Testing (desktop)	Primary Chromium-family browser for desktop verification — including the cross-browser circular-dot rendering check that drove the move from CSS border-radius to inline SVG, plus HTTPS WebSocket gating and reduced-motion behavior.
Android Studio	Testing (mobile)	Android emulator for touch and small-viewport testing — confirming the responsive layout, touch interaction in the Signature view, and rendering of the radial canvas on mobile.

Functional Decisions

What the app does — the behaviors and features that express its single-promise philosophy.

- **One promise at a time —** A single active commitment rather than a list, grounded in the limits of working memory (Cowan's 4±1).
- **Guided entry —** On entering with no promise set for today, the app offers about thirty example promises — a mix of earnest and gently playful — to tap, or fields to write one's own.
- **Break = reflect, not punish —** Missing a promise opens a short reflection rather than a penalty; there are no streaks, scores or shame mechanics anywhere.
- **Five graduated social modes —** Solo, Witness, Reciprocal, Anon and Circle — with the mode being a property of each individual promise, not of the app as a whole. Social presence is always invited, never forced.
- **Quiet Witness —** One trusted person sees a single silent dot and cannot comment, synchronized over a WebSocket relay with a BroadcastChannel fallback for the same device.
- **Living Artifact —** A generative canvas whose vitality reflects the kept rate, paired with an ambient soundscape whose mood follows the app's state.
- **History in two views —** A radial Signature — now at the center, the past in weekly rings outward, angle by weekday, dots marked kept/reflected/missed, with a now–past axis, a zoom control, and tap-to-revisit — alongside a chronological Timeline list.
- **Activity logging with an analytics seam —** A centralised, pluggable logger ready for future analytics, plus a debug overlay that is off by default and shown only with the ?debug=1 URL parameter.
- **Appearance —** Four color palettes, three font pairings, and a motion-intensity control.

Non-Functional Decisions

How the app behaves — the qualities and constraints that hold across every feature.

- **Privacy-first, local-first —** Nothing leaves the device by default; promise content is never logged verbatim (it is reduced to length and word-count metadata); the session identifier is ephemeral; and the Signature is derived client-side so promise content need never leave the device.
- **Calm technology —** Presence is invited rather than forced, motion is minimal and optional, the system honors the operating system's reduced-motion preference, and engagement-maximizing patterns (streaks, scores, fidget mechanics) were deliberately and repeatedly pressure-tested out.
- **Accessibility —** The Timeline list is the canonical, screen-reader-friendly path into one's history; circular dots are drawn with inline SVG because CSS border-radius distorts under flex layouts; reduced motion is respected; controls are keyboard-operable; and canvases carry screen-reader summaries.
- **Performance —** The Living Artifact runs on an OffscreenCanvas worker thread when supported, with an identical-output main-thread fallback; rendering pauses when the tab is hidden or the canvas is off-screen; it is devicePixelRatio-aware and uses GPU-compositing hints.
- **Reliability —** A React error boundary plus global error and unhandled-rejection handlers, with previously silent catch blocks routed through the logger, and a StrictMode-safe teardown for the transferred canvas.

- **Maintainability —** A modular structure (logger, capability detection, artifact renderer/worker / controller, error boundary) delivered as a ready-to-run package, with consistent design tokens across palettes and dark mode.
- **Portability & hosting —** A static build configured for nested GitHub Pages paths (relative base), with HTTPS-aware gating of the WebSocket relay.
- **Security (design exploration) —** TLS for data in motion and encryption at rest; client-side (end-to-end) encryption of promise content; OIDC/PKCE authentication behind a Backend-for-Frontend token pattern; Postgres row-level security scoped to per-promise sharing grants; authenticated WebSocket rooms; and threat modeling via STRIDE and the OWASP ASVS.

Application Navigation Map

Kept is a small, shallow application by design — one welcome screen leads into a single app shell with three bottom-nav destinations and two modal overlays. The map below shows how a person moves through it.

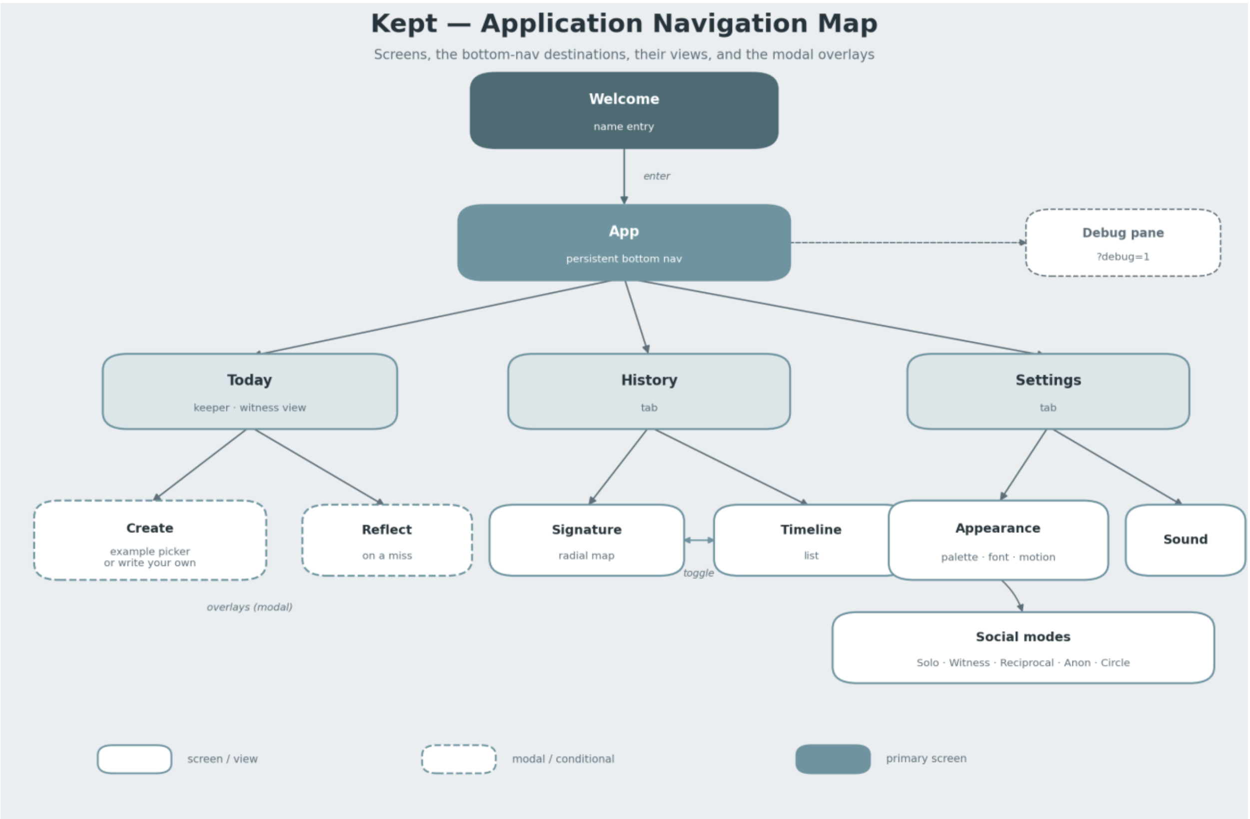

Figure 1. Application navigation map — screens, destinations, views, and overlays.

From the **Welcome** screen the person enters the app shell, whose persistent bottom navigation offers three destinations. **Today** shows the single active promise (the view differs for the keeper and the witness role) and opens the **Create** and **Reflect** overlays. **History** holds the Signature and Timeline views behind a toggle. **Settings** covers appearance, sound, and the five social modes. The debug pane is reachable only via the ?debug=1 URL parameter.

Visual Evolution

The app's central question — how to show someone the shape of their kept promises without turning it into a score — drove the most visible design iteration. The figure below traces that arc, ending in the two History views that coexist in the finished app.

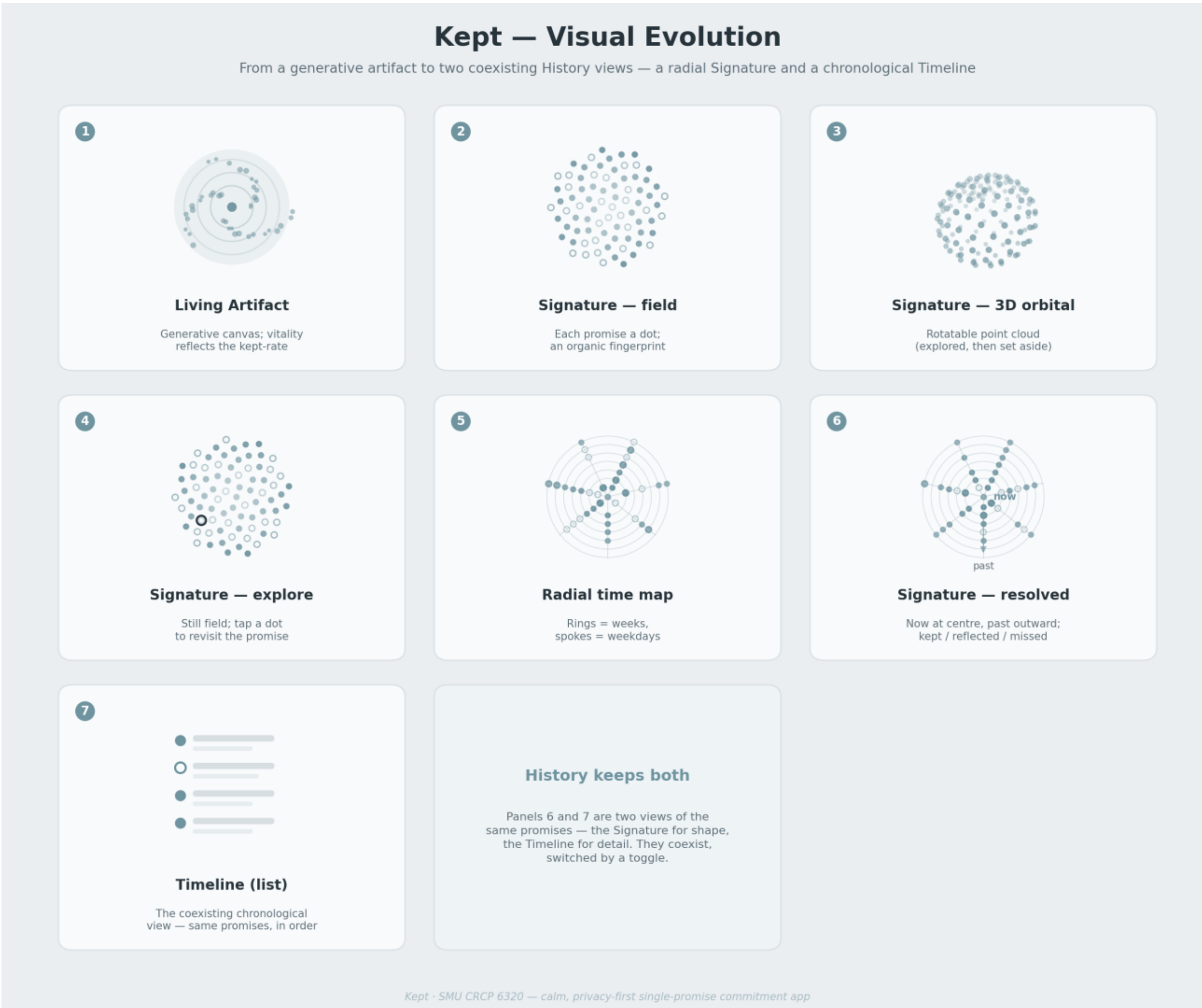

Figure 2. The visual evolution of Kept, from generative artifact to two coexisting History views.

1 — Living Artifact. The app began with a generative canvas whose vitality rose and fell with the kept-rate.

2 — Signature, field. The first attempt at a per-person record placed each promise as a dot in an organic golden-angle field — a unique fingerprint, but not legibly tied to time.

3 — 3D orbital. A rotatable three-dimensional point cloud was explored, then set aside as too toy-like for a calm tool, and because rotation hid meaning behind interaction.

4 — Explore. The field was made still and explorable: hover or tap a dot to revisit the actual promise behind it, shifting the interaction from manipulating a shape to recalling a commitment.

5 — Radial time map. Time was made predictable: concentric rings for weeks and spokes for weekdays. Newest-at-centre and oldest-at-centre layouts were weighed against each other for legibility.

6 — Signature, resolved. The final Signature keeps now at the centre and the past spiralling outward, with labelled weekday spokes, three dot states (kept, reflected, missed), a now–past axis, and tap-to-revisit — a readable record that expresses the kept-rate without ever tallying it.

7 — Timeline. The Signature did not replace the original chronological list. The Timeline coexists with it as the precise, screen-reader-friendly view; the two are switched by a toggle inside History (panels 6 and 7 are two windows onto the same promises).

User-Testing Event Data

Kept's centralised logger records activity as content-free events (names and metadata only — never the text of a promise). To support user testing, two datasets were captured for the same person across two first sessions: one acting as the promise keeper, one as the promise witness. Each dataset is a chronological event log (delivered as a CSV alongside this document); the chart below shows how often each event fired in each session. Not every event fires in a given role — a keeper never receives a witness update, and a witness never creates or keeps a promise — so the two profiles diverge sharply.

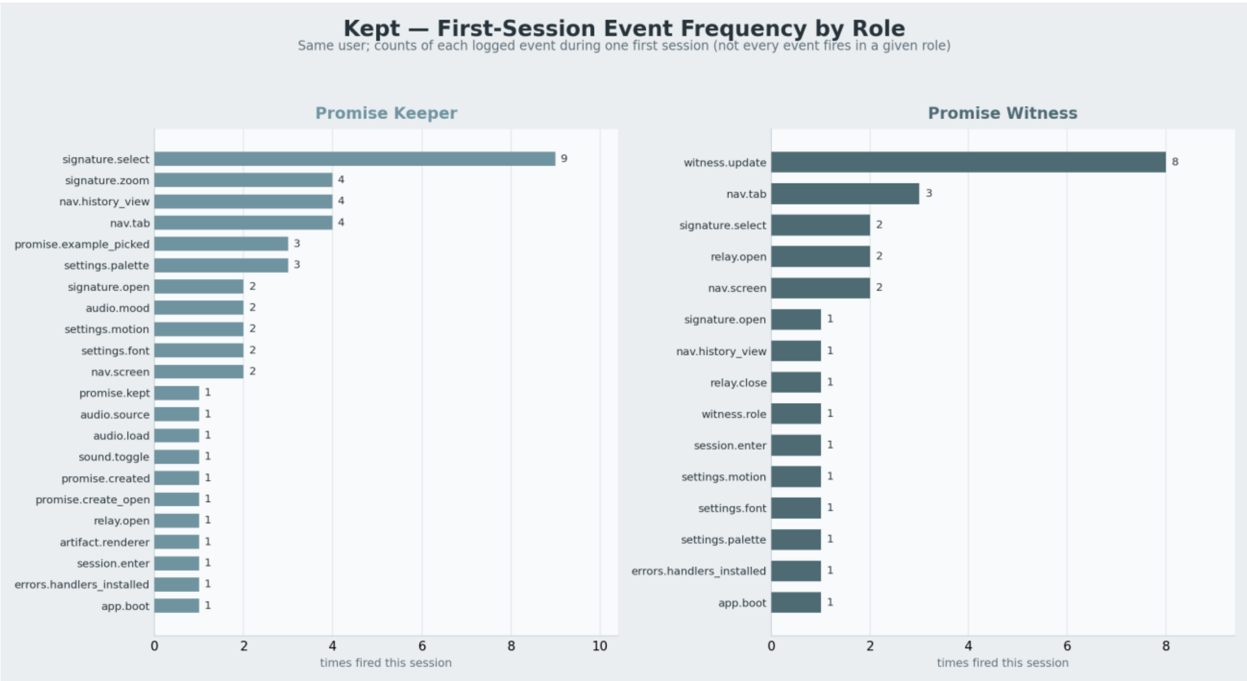

Figure 3. First-session event frequency for the keeper and witness roles.

Keeper. The session is dominated by exploration and authoring: signature.select leads by a wide margin (revisiting individual dots in the Signature), followed by signature.zoom, nav.history_view and nav.tab. The promise lifecycle itself — create_open, created, kept — fires once each, as expected for a single daily promise.

Witness. The same person, witnessing, produces a very different shape: witness.update dominates (the relayed status of the keeper's promise arriving repeatedly), with relay.open and nav.tab next. Authoring events are absent entirely. This contrast is exactly what the roles should produce, and it validates that the logging distinguishes them.

Datasets. keeper_session_events.csv (48 events) and witness_session_events.csv (27 events) accompany this document. Each row carries a sequence number, a relative timestamp, the side, kind, event name, level, and a metadata detail field — mirroring the logger's on-device record shape.